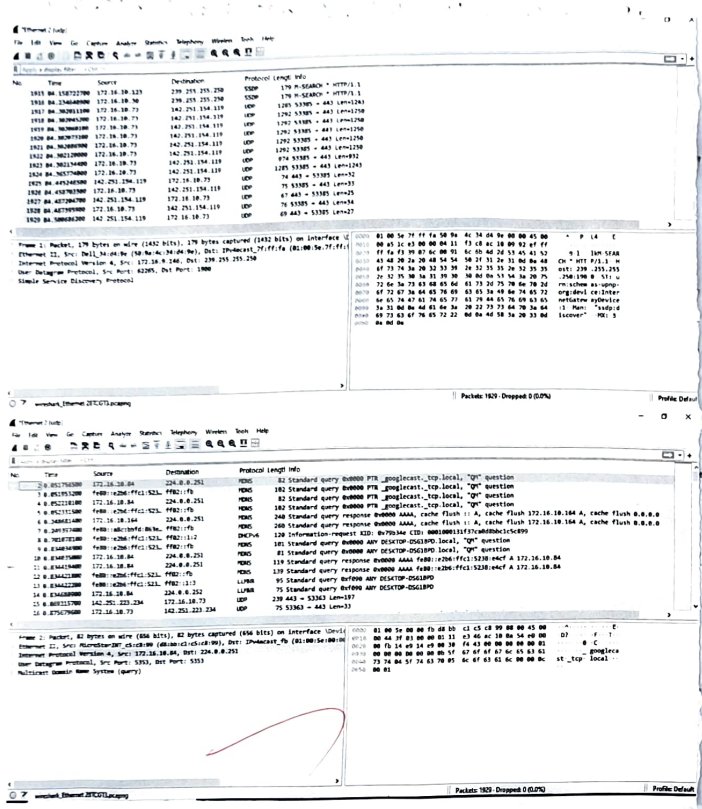
Ex. no: 5

Date: 6.08.26

Experiment on packet capture tool : Wireshark.



* Aim :

To understand the feature of wireshark as a packet capture tool and understand encapsulation of information at various layers of a protocol stack.

* Packet Sniffer.

. Captures packets sent and received by the computer.

. Displays protocol information of captured packets

. Works passively.

. Receives copies of Networks packets for Analysis.

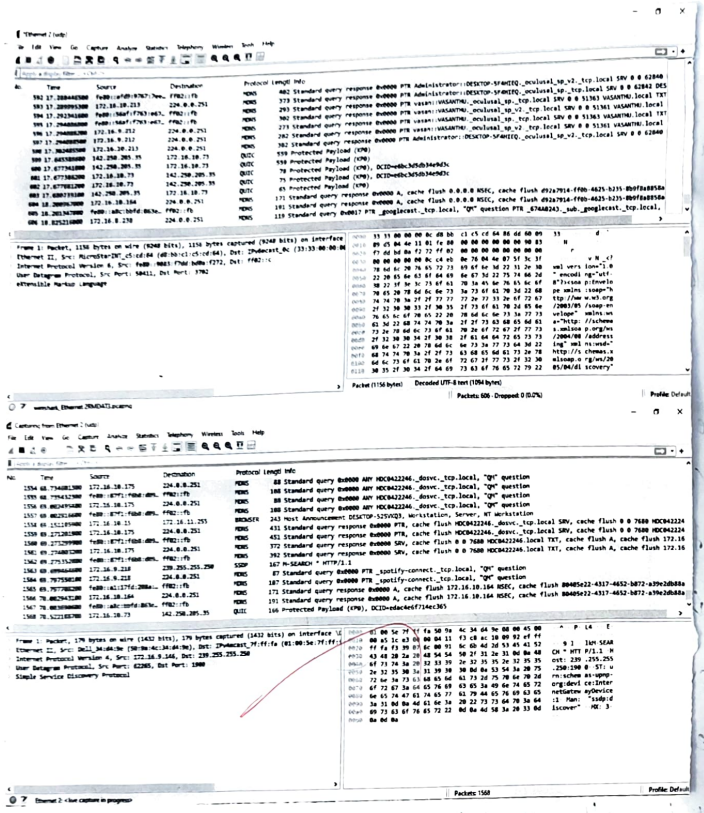
* Packet Sniffer Tools:

. TCP dump - Command line packet capture tool.

. Wire shark - Graphical packet analyzer.

* Description :-

Wireshark is a networks protocols analyzer that captures and displays networks packets in real time. It helps monitor, analyze, troubleshoot and inspect networks traffic using filters and



packet details.

. Uses of Networks:

. Capture networks traffic.
. Analyzers protocols.
. Applies displays filters.
. monitors Networks issues.

* Applications:

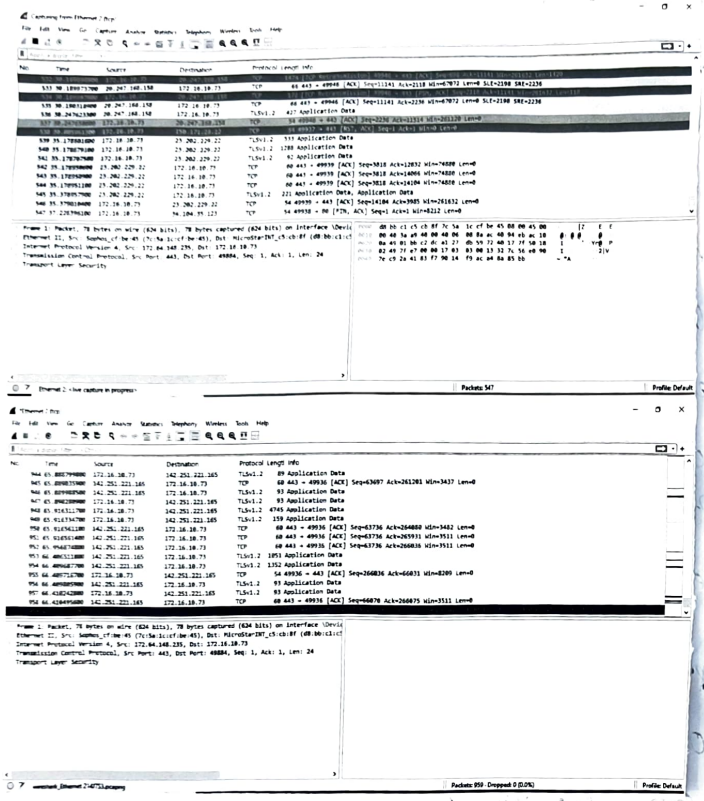
. Network troubleshooting.
. Security analyses.
. Protocol debugging.
. Learning network protocols.

* Capturing packet:

1. Open wireshark.
2. Select the network interface
3. click start Capture.
4. Observes packet in real time.
5. click stop and save the captures.

* Wireshark Panes:

. packet List: Display Captures packets.
. packet Details: Shows protocols information.
. packet Bytes: Display packet data in hexadecimal points.



* Color Coding:
  - purple : TCP
  - Blue : UDP
  - Black : Error packets.
* Filtering Packets:
  - Enter a letter in the filter bar.
  - Right click to apply filters or follow the TCP stream
* Flow Graph:
  Displays the communication flow between devices, making packing exchange easier to understand.
* Capture two packets.
  1. Open wireshark.
  2. Select the network interface.
  3. Set the capture limit into 100 packets.
  4. Start Capturing.
  5. Start and save the captured packets.

Analysing packets using Wireshark tool.

1) Create a filter to Displays only TCP/UDP Packets.
   1. Open the Capture file.

2. Apply the TCP || udp filter.
3. inspect and view the flow graph.

2. Create a file to display only ARP Packets
   1. Open the capture file.
   2. Apply the arp filter.
   3. inspect and view the flow graph

3. Create a filter to display only DNS packets.
   1. Open the capture file.
   2. Apply the DNS filter.
   3. inspect and view the flow graph.

4. Create a filter to display only IP/Icmp packets.
   1. Open the capture file.
   2. Apply the ip or icmp filter.
   3. inspect and view the flow graph and save capture.

5. Create a filter to display only DHCP packet
   1. open the Capture file.
   2. Apply the dhcp file.

5. Inspect and view the flow graph and save the capture.

* Result:

Thus the feature of wireshark and the implement of different filter was successfully completed.